

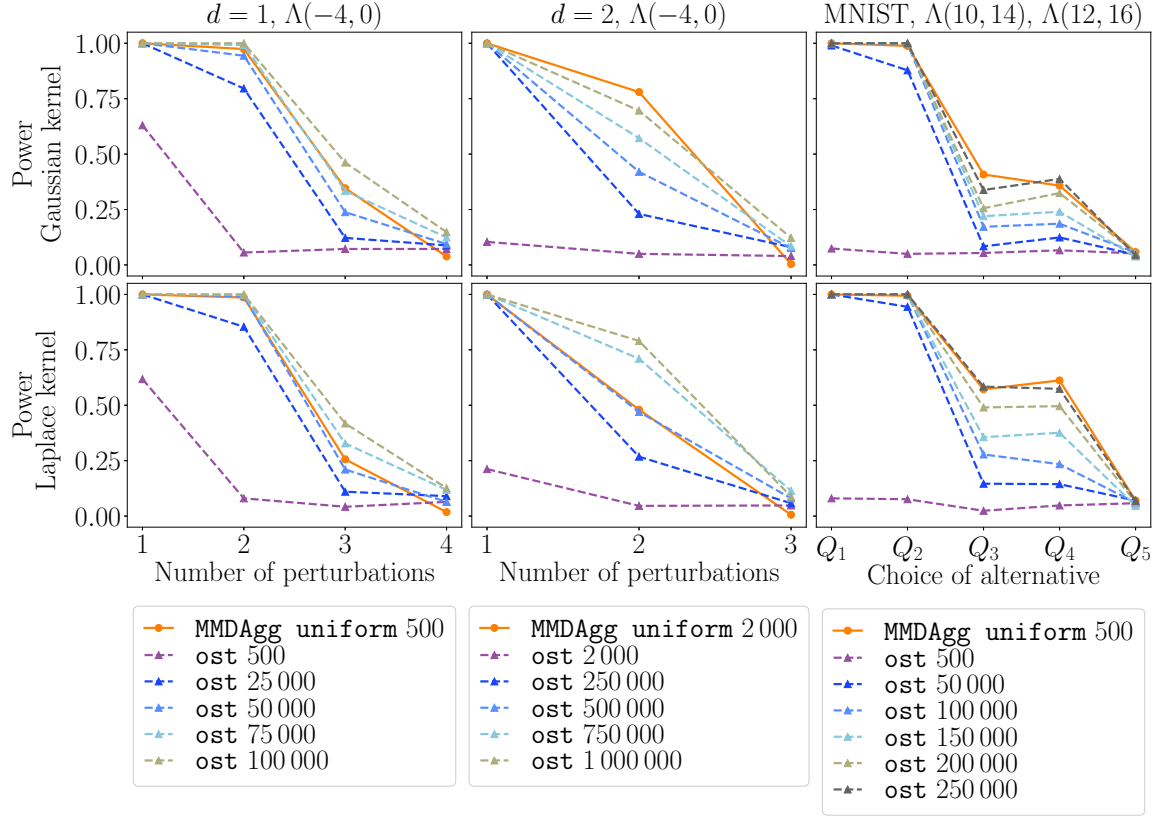


Figure 10: Power experiments increasing the sample sizes for the `ost` test on perturbed uniform d -dimensional distributions and on the MNIST dataset. The legend lists the name of the test followed by the sample sizes used.

is restricted to the use of the linear-time MMD estimate, and hence obtains low power compared to the other tests which use the quadratic-time MMD estimate. We increase the sample sizes for the `ost` test until it matches the power of `MMDAgg uniform` with fixed sample sizes.

For the case of 1-dimensional perturbed uniform distributions, we observe in Figure 10 that `ost` requires sample sizes of 75 000 and 50 000 in order to match the power obtained by `MMDAgg uniform` with $m = n = 500$ for the Gaussian and Laplace kernels, respectively. For the case of 2-dimensional perturbed uniform distributions, one million, and half a million samples are required for the Gaussian and Laplace kernels, respectively, to obtain the same power as `MMDAgg uniform` with 2000 samples. When working with the MNIST dataset, it takes 250 000 samples for `ost` to achieve similar power to the one obtained by `MMDAgg uniform` with 500 samples.

Recall that the MNIST dataset consists of 70 000 images, so it is interesting to see that the power of `ost` keeps increasing for sample sizes which are more than ten times bigger than the size of the dataset. This is due to the use of the linear-time MMD estimate which

for even sample sizes $n = m$ is equal to

$$\frac{2}{n} \sum_{i=1}^{n/2} h_k(X_{2i-1}, X_{2i}, Y_{2i-1}, Y_{2i})$$

for h_k defined as in Equation (5). The two pairs of samples (X_{2i-1}, X_{2i}) and (Y_{2i-1}, Y_{2i}) only appear together as a 4-tuple in this estimate, for $i = 1, \dots, n/2$. So, as long as we do not sample exactly the two same pairs of images together, this creates a new 4-tuple which is considered as new data for this estimate. This explains why considering sample sizes much larger than 70 000 still results in an increase in power.

Appendix B. Relation between permutations and wild bootstrap

In this section, we assume that we have equal sample sizes $m = n$ and show the relation between using permutations and using a wild bootstrap for the estimator $\widehat{\text{MMD}}_{\lambda, \mathbf{b}}^2$ defined in Equation (6).

First, we introduce some notation. For a matrix $A = (a_{i,j})_{1 \leq i,j \leq 2n}$, we denote the sum of all its entries by A_+ and denote by A° the matrix A with all the entries

$$\{a_{i,i}, a_{n+i,n+i}, a_{n+i,i}, a_{i,n+i} : i = 1, \dots, n\}$$

set equal to 0. Note that A is composed of four $(n \times n)$ -submatrices, and that A° is the matrix A with the diagonal entries of those four submatrices set to 0. We let $\mathbb{1}_n \in \mathbb{R}^{n \times 1}$ denote the vector of length n with all entries equal to 1. We also let $v := (\mathbb{1}_n, -\mathbb{1}_n) \in \mathbb{R}^{2n \times 1}$ and note that

$$vv^\top = \begin{pmatrix} \mathbb{1}_n \mathbb{1}_n^\top & -\mathbb{1}_n \mathbb{1}_n^\top \\ -\mathbb{1}_n \mathbb{1}_n^\top & \mathbb{1}_n \mathbb{1}_n^\top \end{pmatrix} \in \mathbb{R}^{2n \times 2n}.$$

We let K_λ denote the kernel matrix $(k_\lambda(U_i, U_j))_{1 \leq i,j \leq 2n}$ where $U_i := X_i$ and $U_{n+i} := Y_i$ for $i = 1, \dots, n$. We let Tr denote the trace operator and $\circ$ denote the Hadamard product.

By definition of Equation (6), we have

$$\begin{aligned} \widehat{\text{MMD}}_{\lambda, \mathbf{b}}^2(\mathbb{X}_n, \mathbb{Y}_n) &:= \frac{1}{n(n-1)} \sum_{1 \leq i \neq j \leq n} h_\lambda(X_i, X_j, Y_i, Y_j) \\ &= \frac{1}{n(n-1)} \sum_{1 \leq i \neq j \leq n} k_\lambda(X_i, X_j) + k_\lambda(Y_i, Y_j) - k_\lambda(X_i, Y_j) - k_\lambda(Y_i, X_j) \\ &= \frac{1}{n(n-1)} \left(K_\lambda^\circ \circ vv^\top \right)_+ \\ &= \frac{1}{n(n-1)} \text{Tr} \left(K_\lambda^\circ vv^\top \right) \\ &= \frac{1}{n(n-1)} \text{Tr} \left(v^\top K_\lambda^\circ v \right) \\ &= \frac{1}{n(n-1)} v^\top K_\lambda^\circ v. \end{aligned}$$

For the wild bootstrap, as presented in Section 3.2.2, we have n i.i.d. Rademacher random variables $\epsilon := (\epsilon_1, \dots, \epsilon_n)$ with values in $\{-1, 1\}^n$, and let $v_\epsilon := (\epsilon, -\epsilon) \in \{-1, 1\}^{2n}$, so that

$$v_\epsilon v_\epsilon^\top = \begin{pmatrix} \epsilon \epsilon^\top & -\epsilon \epsilon^\top \\ -\epsilon \epsilon^\top & \epsilon \epsilon^\top \end{pmatrix} \in \mathbb{R}^{2n \times 2n}.$$

As in Equation (11), we then have

$$\begin{aligned} \widehat{M}_\lambda^\epsilon &:= \frac{1}{n(n-1)} \sum_{1 \leq i \neq j \leq n} \epsilon_i \epsilon_j h_\lambda(X_i, X_j, Y_i, Y_j) \\ &= \frac{1}{n(n-1)} \sum_{1 \leq i \neq j \leq n} \epsilon_i \epsilon_j k_\lambda(X_i, X_j) + \epsilon_i \epsilon_j k_\lambda(Y_i, Y_j) - \epsilon_i \epsilon_j k_\lambda(X_i, Y_j) - \epsilon_i \epsilon_j k_\lambda(Y_i, X_j) \\ &= \frac{1}{n(n-1)} \left(K_\lambda^\circ \circ v_\epsilon v_\epsilon^\top \right)_+ \\ &= \frac{1}{n(n-1)} \text{Tr} \left(K_\lambda^\circ v_\epsilon v_\epsilon^\top \right) \\ &= \frac{1}{n(n-1)} v_\epsilon^\top K_\lambda^\circ v_\epsilon. \end{aligned}$$

We introduce more notation. For a matrix $A = (a_{i,j})_{1 \leq i,j \leq 2n}$ and some permutation $\tau: \{1, \dots, 2n\} \rightarrow \{1, \dots, 2n\}$, we denote by $A^{\circ\tau}$ the matrix $\bar{A}$ with all the entries

$$\{a_{\tau(i), \tau(j)}, a_{\tau(n+i), \tau(n+j)}, a_{\tau(n+i), \tau(j)}, a_{\tau(i), \tau(n+j)} : i = 1, \dots, n\}$$

set to be equal to 0. We denote by A_τ the permuted matrix $(a_{\tau(i), \tau(j)})_{1 \leq i,j \leq 2n}$. Similarly, for a vector $w = (w_1, \dots, w_{2n})$, we write the permuted vector as $w_\tau = (w_{\tau(1)}, \dots, w_{\tau(2n)})$.

Recall that $v = (v_1, \dots, v_{2n}) = (\mathbb{1}_n, -\mathbb{1}_n) \in \mathbb{R}^{2n \times 1}$. Similarly to Equation (10) in Section 3.2.1, but for the estimator $\widehat{\text{MMD}}_{\lambda, \mathbf{b}}^2$, given a permutation $\sigma: \{1, \dots, 2n\} \rightarrow \{1, \dots, 2n\}$, we can define $\widehat{M}_\lambda^\sigma$ as

$$\begin{aligned} &\frac{1}{n(n-1)} \sum_{1 \leq i \neq j \leq n} h_\lambda(U_{\sigma(i)}, U_{\sigma(j)}, U_{\sigma(n+i)}, U_{\sigma(n+j)}) \\ &= \frac{1}{n(n-1)} \sum_{1 \leq i \neq j \leq n} k_\lambda(U_{\sigma(i)}, U_{\sigma(j)}) + k_\lambda(U_{\sigma(n+i)}, U_{\sigma(n+j)}) - k_\lambda(U_{\sigma(i)}, U_{\sigma(n+j)}) - k_\lambda(U_{\sigma(n+i)}, U_{\sigma(j)}) \\ &= \frac{1}{n(n-1)} \left(((K_\lambda)_\sigma)^\circ \circ v v^\top \right)_+ \\ &= \frac{1}{n(n-1)} \left((K_\lambda^{\circ\sigma})_\sigma \circ v v^\top \right)_+ \\ &= \frac{1}{n(n-1)} \left(K_\lambda^{\circ\sigma} \circ (v v^\top)_{\sigma^{-1}} \right)_+ \\ &= \frac{1}{n(n-1)} \text{Tr} \left(K_\lambda^{\circ\sigma} v_{\sigma^{-1}} v_{\sigma^{-1}}^\top \right) \\ &= \frac{1}{n(n-1)} v_{\sigma^{-1}}^\top K_\lambda^{\circ\sigma} v_{\sigma^{-1}}. \end{aligned}$$